

# Phase 3

## Computer Vision

June 2025 Monthly Roadmap | Computer Vision Mastery Program

| Author               | Institute          | Supervisor    | Month     | Phase  |
|----------------------|--------------------|---------------|-----------|--------|
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### ■ Phase 3 Overview

Phase 3 marks the transition from Classical Machine Learning (Phase 2) to Deep Learning. This phase covers the mathematical foundations of neural networks, CNN architectures from scratch, state-of-the-art object detection (YOLO, Faster R-CNN), semantic and instance segmentation, and culminates in Vision Transformers (ViT, Swin, DETR). Every concept is implemented from scratch in PyTorch before using pre-trained weights — aligning with the repository philosophy: *'If you can't implement it from scratch, you don't understand it yet.'*

| Weeks  | Working Days | Daily Schedule        | Projects              | AM Session                    | PM Session                    |
|--------|--------------|-----------------------|-----------------------|-------------------------------|-------------------------------|
| 9 – 14 | 30 days      | 8 hrs / day (Mon–Fri) | 6 Weekly + 1 Capstone | 9:00 AM – 1:00 PM<br>(Theory) | 2:00 PM – 6:00 PM<br>(Coding) |

### ■■ June 2025 — Weekly Breakdown at a Glance

| Week    | Dates<br>(Mon–Fri)           | Theme                               | Key Topics                                                          | Weekly Project                                        |
|---------|------------------------------|-------------------------------------|---------------------------------------------------------------------|-------------------------------------------------------|
| Week 9  | 1 – 5 Jun 2026               | Neural Network Foundations          | MLP math, PyTorch, Autograd, BatchNorm, Dropout, CNN theory         | <i>Custom CNN Classifier</i>                          |
| Week 10 | 8 – 12 Jun 2026              | Classic CNN Architectures           | LeNet→AlexNet→VGG→ResNet<br>Inception, DenseNet, Transfer Learning  | <i>Plant Disease Classifier + Grad-CAM</i>            |
| Week 11 | 15 – 19 Jun 2026             | Object Detection with Deep Learning | Anchor boxes, NMS, R-CNN family,<br>YOLOv1→v8, Custom YOLO training | <i>Real-time Object Detector<br/>(webcam+ONNX)</i>    |
| Week 12 | 22 – 26 Jun 2026             | Semantic & Instance Segmentation    | FCN, U-Net, DeepLabv3, Mask R-CNN,<br>SAM, Panoptic segmentation    | <i>Medical Image Segmentation (U-Net)</i>             |
| Week 13 | 29 – 30 Jun 2026 (cont. Jul) | Attention & Vision Transformers     | Self-attention, Multi-head, ViT,<br>CLIP, positional encoding       | <i>Transformer block from scratch + ViT fine-tune</i> |
| Week 14 | Jul 2026                     | Swin, DETR & Phase 3 Capstone       | Hierarchical ViT, DETR bipartite matching, multi-task pipeline      | <i>Classify + Detect + Segment unified pipeline</i>   |

### ■ Detailed Weekly Plans

#### Week 9 | Neural Network Foundations

June 1 – 5, 2026

**Objective:** Build the mathematical and coding foundation for all deep learning work ahead. Every component — forward pass, backpropagation, optimizers — is implemented from scratch in NumPy and PyTorch before using any high-level API.

| Day | Topic                 | Key Concepts                                                                                                     | Daily Deliverable                             |
|-----|-----------------------|------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| Mon | MLP Theory            | Perceptron model, activation functions (ReLU/Sigmoid/Tanh), forward pass, backpropagation derivation, chain rule | MLP from scratch in NumPy on MNIST            |
| Tue | PyTorch Fundamentals  | Tensors, autograd mechanism, nn.Module structure, Dataset/DataLoader pipelines, GPU transfer                     | MLP on MNIST using PyTorch                    |
| Wed | Optimizers & Training | SGD, Momentum, Adam, LR schedulers, BatchNorm math, Dropout, weight initialization strategies                    | Optimizer comparison study (loss curves)      |
| Thu | CNN Theory            | Convolution math, stride/padding, receptive field, feature map dimensions, pooling layers, parameter count       | CNN from scratch in PyTorch                   |
| Fri | Weekly Project        | Full custom dataset pipeline: data loading, augmentation, training loop, validation, checkpoint saving           | PROJECT: Custom CNN Classifier on own dataset |

**Weekly Project Deliverable:** Custom CNN Classifier — end-to-end training pipeline with accuracy/loss curves and confusion matrix.

## Week 10 | Classic CNN Architectures

June 8 – 12, 2026

**Objective:** Study the evolutionary history of CNN architectures — from LeNet to EfficientNet — understanding the design decisions that led to each breakthrough. Implement ResNet from scratch and master transfer learning for real-world datasets.

| Day | Topic                    | Key Concepts                                                                                           | Daily Deliverable                              |
|-----|--------------------------|--------------------------------------------------------------------------------------------------------|------------------------------------------------|
| Mon | LeNet → AlexNet → VGGNet | Architecture evolution, depth vs width tradeoffs, GPU training impact, data augmentation strategies    | VGGNet on CIFAR-100                            |
| Tue | ResNet                   | Vanishing gradient problem, residual/skip connections, bottleneck blocks, ResNet-18/34/50/101 variants | ResNet-18 from scratch in PyTorch              |
| Wed | Inception & DenseNet     | Parallel convolutions, Inception module, DenseNet dense connections, EfficientNet compound scaling     | Architecture benchmark comparison              |
| Thu | Transfer Learning        | Feature extraction vs fine-tuning, layer freezing strategies, domain adaptation, learning rate warm-up | ResNet-50 fine-tuned on 5-class custom dataset |
| Fri | Weekly Project           | Grad-CAM implementation, class activation maps, visual explainability, model interpretability          | PROJECT: Plant Disease Classifier + Grad-CAM   |

**Weekly Project Deliverable:** Plant Disease Classifier — ResNet-50 fine-tuned with Grad-CAM visualizations showing which leaf regions drive predictions.

## Week 11 | Object Detection with Deep Learning

June 15 – 19, 2026

**Objective:** Master object detection from fundamentals (anchor boxes, NMS) through the R-CNN family to modern YOLO. Build a custom real-time detector exported to ONNX for deployment-ready inference.

| Day | Topic | Key Concepts | Daily Deliverable |
|-----|-------|--------------|-------------------|
|-----|-------|--------------|-------------------|

|     |                          |                                                                                                                   |                                                           |
|-----|--------------------------|-------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| Mon | Detection Fundamentals   | Anchor boxes, IoU calculation, Non-Maximum Suppression (NMS), regression + classification head design, mAP metric | NMS from scratch implementation                           |
| Tue | R-CNN Family             | R-CNN, Fast R-CNN, Faster R-CNN architecture, Region Proposal Network (RPN), ROI Pooling/Align                    | Faster R-CNN inference on COCO                            |
| Wed | YOLO Architecture        | YOLOv1→v8 evolution, grid-based prediction, multi-scale detection heads, anchor-free design                       | YOLOv8 fine-tuning on custom dataset                      |
| Thu | Custom Training Pipeline | LabelImg annotation, YOLO label format, Albumentations augmentation, training on 200+ images                      | 200-image custom object detector                          |
| Fri | Weekly Project           | ONNX model export, OpenCV DNN inference, webcam real-time pipeline, FPS optimization                              | PROJECT: Real-time Object Detector (webcam + ONNX export) |

**Weekly Project Deliverable:** Real-time Object Detector — YOLOv8 trained on custom class, exported to ONNX, running live on webcam feed.

## Week 12 | Semantic & Instance Segmentation

June 22 – 26, 2026

**Objective:** Go beyond bounding boxes to pixel-level understanding. Implement U-Net from scratch, study DeepLab atrous convolutions, use Mask R-CNN for instance segmentation, and explore the Segment Anything Model (SAM) for zero-shot tasks.

| Day | Topic                       | Key Concepts                                                                                                  | Daily Deliverable                                             |
|-----|-----------------------------|---------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| Mon | FCN & U-Net                 | Fully Convolutional Networks, encoder-decoder architecture, skip connections, U-Net implementation, Dice loss | U-Net from scratch on Oxford-IIIT Pets                        |
| Tue | DeepLab Family              | Atrous (dilated) convolution, Atrous Spatial Pyramid Pooling (ASPP), DeepLabv3+ decoder, CRF post-processing  | DeepLabv3+ on Cityscapes subset                               |
| Wed | Mask R-CNN                  | ROI Align (vs ROI Pooling), mask prediction branch, instance vs semantic segmentation, Detectron2 setup       | Mask R-CNN via Detectron2                                     |
| Thu | SAM & Panoptic Segmentation | Segment Anything Model architecture, prompt types (point/box/text), zero-shot segmentation, panoptic FPN      | SAM vs fine-tuned U-Net comparison                            |
| Fri | Weekly Project              | Medical image segmentation, Dice/IoU metrics, data augmentation for medical domain, model evaluation          | PROJECT: Medical Image Segmentation (U-Net + full evaluation) |

**Weekly Project Deliverable:** Medical Image Segmentation — U-Net trained on medical dataset with Dice score, IoU metrics, and visual overlay results.

## ■ Weeks 13–14 — Preview (Late June / Early July)

Weeks 13 and 14 extend Phase 3 into Vision Transformers and serve as the bridge to Phase 4 (Advanced Deep Learning). They are included in the June plan as the 6-month roadmap schedules them for the final two weeks of the June–July boundary.

| Week    | Core Theme                      | Topics                                                                                                                                       | Deliverable                                                       |
|---------|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| Week 13 | Attention & Vision Transformers | Self-attention derivation, multi-head attention, positional encoding, ViT patch embedding, CLIP contrastive pretraining, zero-shot inference | Transformer block from scratch + ViT fine-tuned on custom dataset |

|                |                                   |                                                                                                                                    |                                                                |
|----------------|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| <b>Week 14</b> | Swin Transformer, DETR & Capstone | Hierarchical ViT, shifted window attention, DETR bipartite matching loss, end-to-end detection, multi-task unified pipeline design | PHASE 3 CAPSTONE: Classify + Detect + Segment unified pipeline |
|----------------|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|

GitHub Repository: [github.com/FahimShah651/CV\\_Practice](https://github.com/FahimShah651/CV_Practice) | Phase 3 folder: **03\_June/** | Supervisor: **Dr. Yasir Jan** | Generated: June 2026